Ryerson University

Department of Mathematics

Sample Test # 1

MTH 240 – Calculus II

Student Number: _____. Last Name:_____. First Name (**Print**):_____.

Signature:_____.

Professor (circle one)

Dr Bonato	Dr Ha
Dr Olivares	Dr Tasić

Section : _____

Date: February 13, 2009, 4 pm

Duration: 1hr. 30 min.

Instructions:

1. Have your student card available on your desk.
2. This is a closed-book test. **Notes, calculators and other aids are not permitted.** Verify that your test booklet has pages 1-6.
3. Do not separate the pages of this test booklet.
4. The point value of each question is indicated by the question number.
5. Include all significant steps in your solutions to the questions, presented in the correct order. **Unjustified answers will be given little or no credit.** Cross out or erase all rough work not relevant to your solution.
6. Present your solutions neatly and legibly in the space provided. **Messy or illegible solutions will receive no credit**.
7. If you need more space, use the back of the previous page. Indicate this fact on the original page.

For Instructor's use only.

Question	Mark
1	
2	
3	
4	
5	
Total	/50

1. [(5 + 5) marks]

(a) Evaluate

$$\int \arctan x \, dx.$$

Put $u = \arctan x$, $dx = dv$. Then $du = \frac{dx}{1+x^2}$ and $x = V$

$$\int \arctan x \, dx = x \arctan x - \underbrace{\int \frac{x}{1+x^2} dx}_{\circledast} = x \arctan x - \frac{1}{2} \ln(1+x^2) + C$$

$\circledast$ $\int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{dy}{y} = \frac{1}{2} \ln|y| = \frac{1}{2} \ln(1+x^2)$

$1+x^2 = y \Rightarrow 2x dx = dy$

(b) Calculate the limit and justify your answer.

$$\lim_{x \to 0^+} (1+4x)^{\cot x}$$

$$\lim_{x \to 0^+} (1+4x)^{\cot x} \overset{1^\infty}{=} \lim_{x \to 0^+} e^{\ln(1+4x)^{\cot x}} = \lim_{x \to 0^+} e^{\cot x \cdot \ln(1+4x)} =$$

$$= e^{\lim_{x \to 0^+} \cot x \cdot \ln(1+4x)} = e^4$$

$\circledast$ $\lim_{x \to 0^+} \cot x \cdot \ln(1+4x) \overset{\infty \cdot 0}{=} \lim_{x \to 0^+} \frac{\ln(1+4x)}{\tan x} \underset{0}{\overset{0}{=}} \lim_{x \to 0^+} \frac{\frac{4}{1+4x}}{\sec^2 x} = 4$

2. [10 marks] Evaluate

$$\int_0^{\pi} \cos^2(2x)\sin^4(2x)\,dx.$$

$$\int_0^{\pi} \cos^2(2x)\sin^4(2x)\,dx = \int_0^{\pi}(\cos 2x \sin 2x)^2 \sin^2 2x\,dx = \int_0^{\pi}\left(\frac{1}{2}\sin 4x\right)^2 \sin^2 2x\,dx =$$

$$= \frac{1}{4}\int_0^{\pi} \sin^2 4x \sin^2 2x\,dx =$$

Using $\sin^2 2x = \frac{1-\cos 4x}{2}$ and $\sin^2 4x = \frac{1-\cos 8x}{2}$

$$= \frac{1}{4}\int_0^{\pi} \frac{1-\cos 8x}{2}\cdot\frac{1-\cos 4x}{2}\,dx = \frac{1}{16}\int_0^{\pi}(1-\cos 4x - \cos 8x + \cos 4x \cos 8x)\,dx =$$

$$= \frac{1}{16}\int_0^{\pi} dx - \frac{1}{16}\int_0^{\pi}\cos 4x\,dx - \frac{1}{16}\int_0^{\pi}\cos 8x\,dx + \frac{1}{16}\int_0^{\pi}\cos 4x \cos 8x\,dx$$

Using $\int \cos \alpha x\,dx = \frac{1}{\alpha}\sin \alpha x$ and $\cos 4x \cos 8x = \frac{1}{2}[\cos 12x + \cos 4x]$

$* \cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$

$$= \frac{1}{16}(\pi - 0) - \frac{1}{16}\cdot\frac{1}{4}(\sin 4\pi - \sin 0) - \frac{1}{16}\cdot\frac{1}{8}(\sin 8\pi - \sin 0) + \frac{1}{32}\int_0^{\pi}\cos 12\,dx + \frac{1}{32}\int_0^{\pi}\cos 4x\,dx$$

$$= \frac{\pi}{16} + \frac{1}{32}\cdot\frac{1}{12}(\sin 12\pi - \sin 0) + \frac{1}{32}\cdot\frac{1}{4}(\sin 4\pi - \sin 0) = \frac{\pi}{16}$$

3. [10 marks] Evaluate

$$\int_{2\sqrt{2}}^{4} \frac{dx}{x^4\sqrt{x^2-4}}.$$

Put $x = 2\sec\theta \Rightarrow dx = 2\sec\theta\tan\theta\, d\theta$ $\quad 2\sec\theta = 2\sqrt{2} \Rightarrow \sec\theta = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$

$0 \leqslant \theta < \frac{\pi}{2}$ $\quad 2\sec\theta = 4 \Rightarrow \sec\theta = 2 \Rightarrow \theta = \frac{\pi}{3}$

So $\displaystyle\int_{2\sqrt{2}}^{4} \frac{dx}{x^4\sqrt{x^2-4}} = \int_{\pi/4}^{\pi/3} \frac{\cancel{2}\sec\theta\,\cancel{\tan\theta}}{16\sec^4\theta\cdot\cancel{2}\cancel{\tan\theta}}\,d\theta = \frac{1}{16}\int_{\pi/4}^{\pi/3} \frac{d\theta}{\sec^3\theta} = \frac{1}{16}\int_{\pi/4}^{\pi/3} \cos^3\theta\, d\theta =$

$\displaystyle = \frac{1}{16}\int_{\pi/4}^{\pi/3} (1-\sin^2\theta)\cos\theta\, d\theta = \frac{1}{16}\int_{\sqrt{2}/2}^{\sqrt{3}/2} (1-u^2)\, du = \frac{1}{16}\left(u - \frac{u^3}{3}\right)\Big|_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} =$

Put $\sin\theta = u$ $\quad \langle \pi/4, \pi/3 \rangle \xmapsto{\sin\theta = u} \langle \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2} \rangle$

$\cos\theta\, d\theta = du$

$$= \frac{1}{16}\left[\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} - \frac{1}{3}\left(\frac{3\sqrt{3}}{8} - \frac{2\sqrt{2}}{8}\right)\right] = \frac{1}{16}\left[\frac{\sqrt{3}-\sqrt{2}}{2} - \frac{3\sqrt{3}-2\sqrt{2}}{24}\right] = \frac{1}{16}\,\frac{12\sqrt{3}-12\sqrt{2}-3\sqrt{3}+2\sqrt{2}}{24} = \frac{9\sqrt{3}-10\sqrt{2}}{24}\cdot\frac{1}{16}$$

4. [10 marks] Evaluate

$$\int \frac{x^5 + 6x^3 - x^2 + 3x - 3}{x^4 + 3x^2}\, dx.$$

First use long division $\underline{x^5+6x^3-x^2+3x-3} : x^4+3x^2 = x$

$\underline{-x^5-3x^3}$

$3x^3-x^2+3x-3$

So $\dfrac{x^5+6x^3-x^2+3x-3}{x^4+3x^2} = x + \dfrac{3x^3-x^2+3x-3}{x^4+3x^2}$.

$\dfrac{3x^3-x^2+3x-3}{x^2(x^2+3)} = \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{Cx+D}{x^2+3} \Rightarrow \cdots (A+C)x^3+(B+D)x^2+3Ax+3B = 3x^3-x^2+3x-3$

$\Rightarrow A+C=3,\ B+D=-1,\ 3A=3,\ 3B=-3 \Rightarrow A=1,\ B=-1,\ C=2,\ D=0$

Finally $\displaystyle\int \frac{x^5+6x^3-x^2+3x-3}{x^4+3x^2}dx = \int\left(x + \frac{1}{x} - \frac{1}{x^2} + \frac{2x}{x^2+3}\right)dx =$

$= \displaystyle\int x\,dx + \int \frac{dx}{x} - \int x^{-2}dx + \int \frac{2x}{x^2+3}dx = \frac{x^2}{2} + \ln|x| + \frac{1}{x} + \ln(x^2+3) + C$

5. [10 marks] Determine if the improper integral below is convergent or divergent. If divergent, justify your answer; if convergent, calculate the value of the integral.

$$\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx$$

$$\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx = \lim_{t\to 0^+} \int_t^1 \frac{\ln x}{\sqrt{x}}\,dx \underset{\circledast}{=} \lim_{t\to 0^+} \left(\cancelto{0}{-2\sqrt{t}\ln t} - 4 + \cancelto{0}{4\sqrt{t}}\right) = -4$$

$\circledast\circledast$

$$\circledast \int_t^1 \frac{\ln x}{\sqrt{x}}\,dx = 2\sqrt{x}\ln x\Big|_t^1 - \int_t^1 \frac{2\sqrt{x}}{x}\,dx = -2\sqrt{t}\ln t - 2\int_t^1 x^{-\frac{1}{2}}dx = -2\sqrt{t}\ln t - 4\sqrt{x}\Big|_t^1$$

$$= -2\sqrt{t}\ln t - 4(1-\sqrt{t})$$

Put $\ln x = u$, $\frac{dx}{\sqrt{x}} = dv \Rightarrow \frac{dx}{x} = du$, $v = 2\sqrt{x}$

$$\circledast\circledast \lim_{t\to 0^+} \sqrt{t}\ln t \overset{0\cdot(-\infty)}{=} \lim_{t\to 0^+} \frac{\ln t}{\frac{1}{\sqrt{t}}} = \lim_{t\to 0^+} \frac{\ln t}{t^{-\frac{1}{2}}} \overset{\frac{-\infty}{\infty}}{=} \lim_{t\to 0^+} \frac{\frac{1}{t}}{-\frac{1}{2}t^{-\frac{3}{2}}} = -2\lim_{t\to 0^+}\sqrt{t} = 0$$